

LUCKY SAH

MECHATRONICS ENGINEERING / ROBOTICS HARDWARE / PROTOTYPING

Kent, OH • Isah@kent.edu • [FILLER: phone] [FILLER: LinkedIn or portfolio URL]

PROFILE

Mechatronics Engineering undergraduate building robotic systems from parametric CAD through fabrication, actuation integration, and embedded hardware. Experience spans mobile robots, expressive mechanisms, modular AI companion platforms, and automated machinery in research and product-development settings.

TECHNICAL TOOLBOX

CAD and mechanical

Autodesk Inventor Professional, Fusion 360, SolidWorks, parametric assemblies, 3D modeling, BOM development

Robotics and integration

Actuation systems, sensors, motor drivers, mechanical-electrical integration, ROS

Embedded and controls

Arduino, microcontrollers, sensor integration, PLC fundamentals, industrial automation

Fabrication

FDM and resin 3D printing, CNC machining, soldering, power tools, silicone, fiber, and polyurethane molding

EXPERIENCE

Robotics Hardware Intern (Co-op)

[FILLER: start–end dates]

Open Drive LLC | Plaipin Product Line

- Translate CAD designs into functional robotic hardware, prototypes, and integrated mechanical subsystems.
- Support mechanical design, assembly, actuation integration, and iterative prototype development for a modular AI companion robot platform.
- **FILLER** – Add one measured result: prototype iterations, fabricated parts, test cycles, or a time/cost improvement.

Prototyping Manager

[FILLER: start]–Present

Advanced Telerobotics Research Lab | Kent State University

- Lead mechanical prototyping for robotics research from parametric CAD through fabrication, assembly, and functional hardware.
- Design and integrate structures, mounts, actuation components, sensors, and embedded hardware for robotic systems.
- Coordinate equipment, workspace, and prototyping support across interdisciplinary research teams.
- **FILLER** – Quantify scope: number of projects, researchers supported, machines maintained, or prototypes delivered.

Automation and Product Development Intern

2022–2024

Thoplo Machine Pvt. Ltd. & Baliyo Ventures Pvt. Ltd.

- Contributed to the design, fabrication, assembly, and testing of automated production machinery and mechanical systems.
- Supported PLC lab setup, wiring, industrial equipment assembly, machining, fiber molding, and production-oriented fabrication.
- **FILLER** – Add a concrete outcome: equipment commissioned, assemblies completed, or process throughput improved.

SELECTED ROBOTICS PROJECTS

Plaipin – Modular AI Companion Robot Platform Led mechanical CAD design of the core device and modular body configurations; developed assemblies integrating actuation, sensors, and onboard hardware.

Mini TurtleBot – Low-Cost Modular Mobile Robot Designed the chassis and motion architecture for low-cost fabrication, modular expansion, and robotic hardware integration.

KAMI Head – Human-Robot Interaction Module Designed an expressive robotic head assembly with mechanical motion systems and embedded components for human-robot interaction.

FILLER – Add a repository, demo, publication, or competition link to the strongest project.

EDUCATION

B.S. Mechatronics Engineering; Minor in Computer Science

Expected 2028

Kent State University | Kent, OH

Focus: Robotics, mechanical systems, electronics, prototyping, and applied engineering design. [FILLER: GPA, honors, or 2–3 advanced courses if strong]

LEADERSHIP AND TECHNICAL ACTIVITIES

Founder and Organizer, Young Scientist Community RECAST, Tribhuvan University Led robotics education, technical workshops, STEM events, coordination, and outreach.

Selected competitions and conferences: World Robot Olympiad (Thailand), FIRST Global Challenge (Switzerland), Asian Development Bank Hackathon, Robo-Battle Bots, Robo Soccer, Founder's Inc (San Francisco), and Asian Pacific Conference of Young Scientists.